



## Department of Computer Science and Engineering

Academic Year 2023– 2024 (Odd Semester)

Degree, Semester & Branch: I Semester B.E CSE A

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Dr.M.Swarna Sudha , AP(SG)/CSE

### Innovative Practice Description

Unit / Topic: Unit II / Operators

Course Outcome: CO2

Topic Learning Outcome: TLO5

Activity Chosen: Crossword Puzzle

Date of implementation : 27.10.2023

- **Justification:**

- Crossword puzzles are word puzzles that often have squares that are shaded in black and white. The objective is to fill the white squares with letters to construct words or phrases by understanding clues that lead to the solutions.
- Operators are the important concepts used in python. It is important to know about the operations and properties of each operator.
- The act of solving crossword puzzles involves recalling information and applying it in different contexts. This process can contribute to better memory retention of the Python operators and their functionalities.
- So, a crossword puzzle activity is chosen to have a better understanding of the operator's concept by finding the suitable word according to the question

- **Time Allotted for the Activity: 15 Minutes**

- **Details of the Implementation:**

- Crossword puzzle of 11 questions were given to each students.
- The students were understanding the questions, patterns and completed the puzzle as shown in Fig 1
- After completion of the puzzle activity, the students were discussed among them and finally the faculty member discussed the answer to make the students aware of the correct answer as shown in Fig 2

- **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO9 | PO10 | PSO1 |
|-----|-----|-----|-----|-----|------|------|
| CO2 | 3   | 1   | 1   | 1   | 1    | 1    |

(1 – Low      2 – Moderate      3 – High)

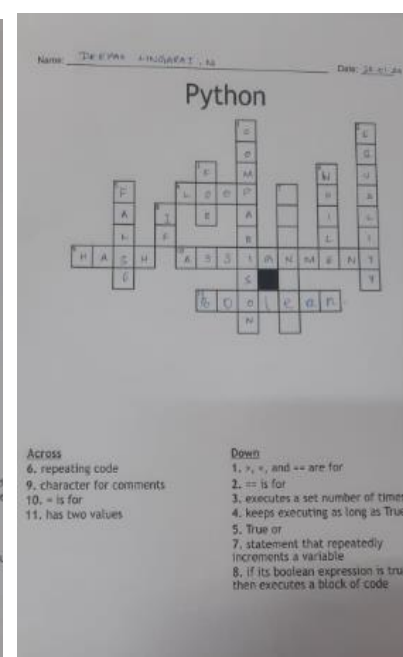
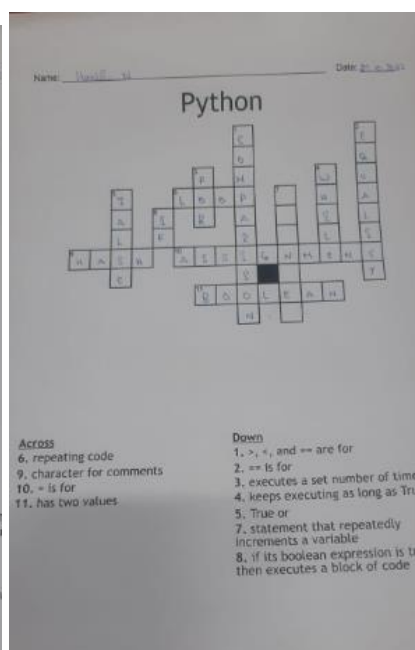
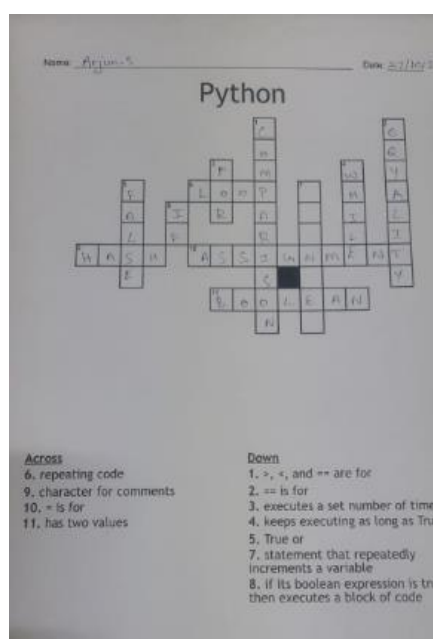
- **PO / PSO mapped:**

| Innovative practice                  | PO1                                                                 | PO2                                                                                  | PO3                                                                                        | PO9                                                                                    | PO10                                                                                        | PSO1                                                                                                  |
|--------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Justification for correlation</b> | Students will be able to understand the concept of python operators | Students will be able to choose the appropriate operator for approaching the problem | Students will be able to provide solution for the problem by applying concept of operator. | As the student involved in the activity as a team of two, the team work skill improved | Students communication skill will be improved as they are discussing the answers with peers | The problem solving skill earned through this activity helps the students in solving various problems |

- **Images / Screenshot of the practice:**



**Fig 1: Students Solving Crossword Puzzle**



ARJUN S

DILIBAN S M

DEEPAK LINGARAJ M

**Fig 2: Sample Sheet of Solved Crossword Puzzle**

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students said that the crossword puzzle activity was very interesting and they were able to identify the appropriate concepts related to list, tuple and dictionary.
- They said the it will be helpful while preparing for examination
- Some of the students find very easy to identify the type of operator.

❖ **Benefit of the practice:**

- Student's understanding of concepts of operators is improved.
- Vocabulary of the terms related to operators improved
- This activity helps to test the level of understanding of the students

❖ **Challenges faced in implementation:**

- Few students found it difficult to complete the puzzle
- Planned for 15 minutes, but unable to be completed within 15 minutes, it will take an extra 5 minutes to complete.
- Few of them were struggled to find the answer for the questions
- Some of the students couldn't able to recall the important terms and definitions while doing the activity. At the end of the class the concept of function revised once again.

**References:**

- <https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-GM.pdf>
- <http://www.classtools.net/crossword/>
- <https://www.activityvillage.co.uk/crosswords>
- [https://www.ritrjpm.ac.in/images/computer-science/2022-2023/BV\\_CWP\\_22-23\\_UnitIV.pdf](https://www.ritrjpm.ac.in/images/computer-science/2022-2023/BV_CWP_22-23_UnitIV.pdf)